

PROJECT NOTES – ONBOARDING ANALYTICS (WORKING NOTES)

Data Model Decisions

Accepted Dimensions

- dim_cliente
- dim_etapa
- dim_fecha
- dim_canal
- dim_riesgo

Fact Table

- fact_onboarding_eventos

Decision:

- No second fact table.
- No fact_onboarding summary table.
- Aggregations will be calculated in SQL views or Power BI.

Canal Dimension

Original version contained customer fields.

Corrected structure:

```
canal_id  
canal_nombre
```

Current channels:

```
Web  
Mobile App  
Branch
```

Decision:

- Do not add Partner for now.

- Avoid scope creep.

Risk Dimension

Created:

dim_riesgo

Values:

Low
Medium
High

Purpose:

- Risk vs Approval Rate
- Risk vs Rejection Rate
- Risk vs Onboarding Time

Document Quality

Current values:

Complete
Incomplete
Expired
Suspicious

Replaced previous:

OK
ERROR

Business interpretation:

Complete

- High approval rate

- Fast onboarding

Incomplete

- Additional document requests
- Longer onboarding

Expired

- Delays
- Higher rejection probability

Suspicious

- Potential fraud indicators
- Highest business risk

Decision:

- No dim_documentacion table.
- Keep estado_documentacion in fact table.

Reason:

- Low business value from separate dimension.
- Faster project completion.

AML Logic

Original field:

aml_flag

Observation:

Current dataset behaves as:

```
aml_flag = 1  
→ Rejected  
  
aml_flag = 0  
→ Approved
```

Business discussion:

This is unrealistic for sanctions screening.

However:

If interpreted as:

AML Risk Score

or

High AML Risk Indicator

the logic becomes acceptable.

Potential rename:

aml_risk_score

Recommended final description:

"AML Risk Score is a synthetic indicator representing the output of an automated AML risk assessment engine based on customer profile, document validation, adverse media signals and onboarding inconsistencies."

Decision:

- Keep current logic.
- Consider renaming aml_flag → aml_risk_score.

Business Interpretation of AML Score

Examples:

High score generated when:

- Business not found online
- Website missing
- Address mismatch
- Email mismatch
- High risk geography
- Suspicious customer activity

- Inconsistent onboarding information
- Fraud indicators

These are treated as:

High AML Risk

and may lead to rejection.

Rejection Drivers (Business Story)

Main rejection drivers:

1. Document Quality
2. Risk Level
3. Business Verification Issues
4. Identity Inconsistencies
5. High AML Risk Score

AML should be interpreted as:

High AML Risk

not:

Sanctions Match

Data Cleaning Activities (Limpieza)

Completed:

Risk Dimension

- Created dim_riesgo
- Added risk_id to fact table

Document Status

Converted:

OK
ERROR

to

Complete
Incomplete
Expired
Suspicious

Canal Dimension

Removed customer-related columns.

Portfolio Narrative

Python

↓

Synthetic dataset generation

↓

PostgreSQL

↓

Data cleaning and normalization

↓

SQL analysis

↓

Power BI

↓

Interactive dashboard



Business insights

Current Priority

1. Validate Risk distribution.
2. Validate Verification Type distribution.
3. Validate Risk vs Approval.
4. Connect dim_fecha.
5. Build Power BI dashboard.

No additional dimensions unless absolutely necessary.